12.4.5 Performance Analysis and Final Considerations

Hierarchical clustering is powerful, but also resource consuming, compared to flat clustering. While it is estimated, in the original paper, that the core OPTICS algorithm runs approximately 1.6 times slower than DBSCAN (on the same datasets), keeping the hierarchical clustering and building and exploring the dendrogram obviously also require extra memory and computation.

For the core algorithm, a quick glance at listing 12.6 and 12.7 shows us that the code processes each point exactly once (because each point is added to and removed from the priority queue once), and runs one region-query for each point processed; however, entries in the priority queue can be updated multiple times, potentially each time a point is processed; the size of the priority queue depends on the size of the dense regions, and the larger is ε_{MAX}, the more points will be in the queue: potentially, the queue could contain all the points starting from iteration 1 (if $\varepsilon_{MAX} \geq$ max pairwise distance), and all points in the queue could be updated each time a point is processed. Likewise, the time needed for nearest neighbor searches, even if using worst-case bounded structures like k-d trees, depends on ε_{MAX}, and if the radius of the region searched is large enough, these queries become linear scans of the dataset.

For these reasons, the worst-case running time of the algorithm is quadratic! Nevertheless, it can be shown that, with an appropriate choice for ε_{MAX}, the nearest neighbor search can be performed in amortized logarithmic time, and